Señales y Sistemas

Guía Prácticos 1

Ing Leandro Borgnino *Teoría de Señales y Sistemas Lineales*

Tema 1: Números Complejos

Calcular las operaciones con números complejos descritas en los ejercicios. Graficar cada uno de los operandos y el resultado.

(1) $z_3 = z_1 \cdot z_2 \quad con \quad z_1 = ej\frac{\pi}{4}\ ,\ z_2 = 1 + j$

$z_3 = \sqrt{2}e^{j\pi/2}$

(2) $z_3 = z_1 + z_2 \quad con \quad z_1 = 5 + 3j\ ,\ z_2 = 3e^{j\frac{2\pi}{3}}.$

$z_3 = 3.5 + j5.6$

(3) $z_3 = (z_1 \cdot z_2) + z_3 \quad con \quad z_1 = 2 + j\ ,\ z_2 = 3e^{j\frac{7\pi}{4}}\ ,\ z_3 = 1 - 5j.$

$z_3 = 7.36 - j7.12$

(4) $z_2 = z_1 + z_1^* \quad con \quad z_1 = 3e^{j\frac{3\pi}{4}}$

$z_3 = -\frac{3}{2}$

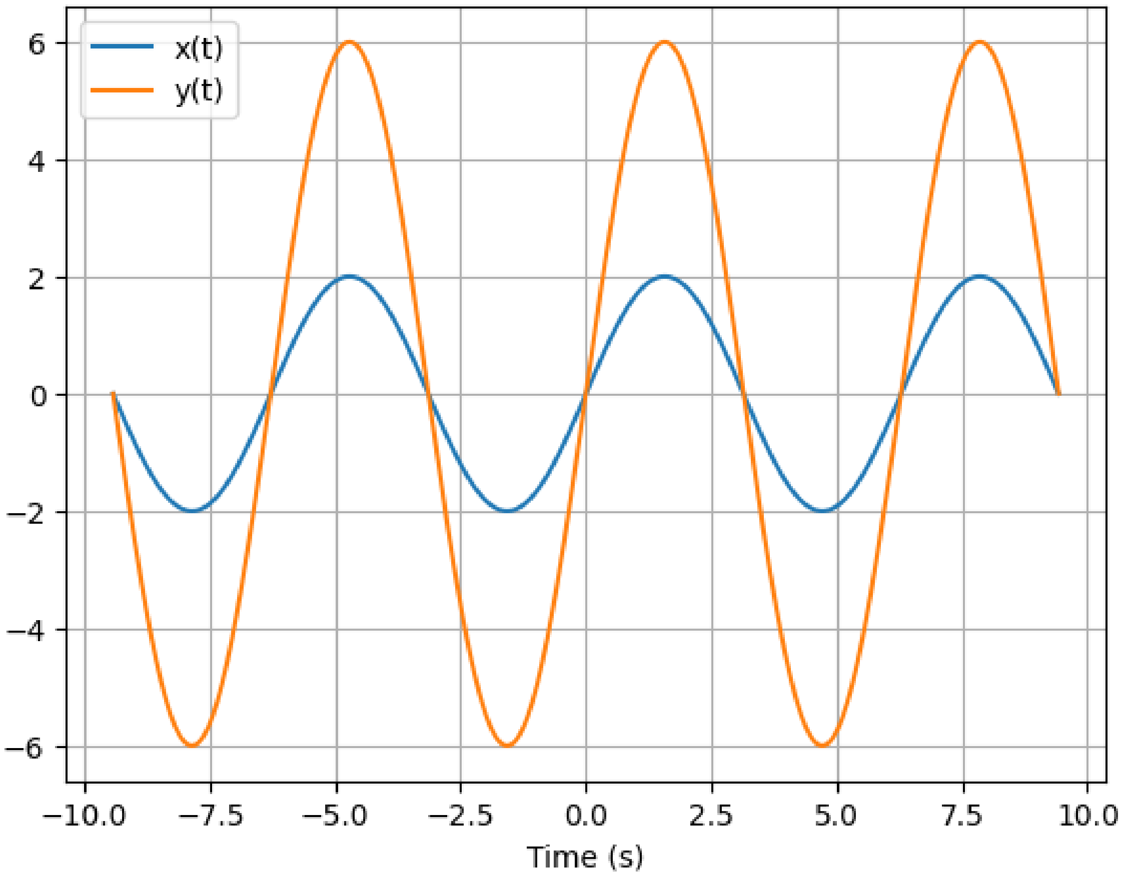
(5) $z_2 = z_1 - z_1^* \quad con \quad z_1 = 2 + j5$

$z_3 = j10$

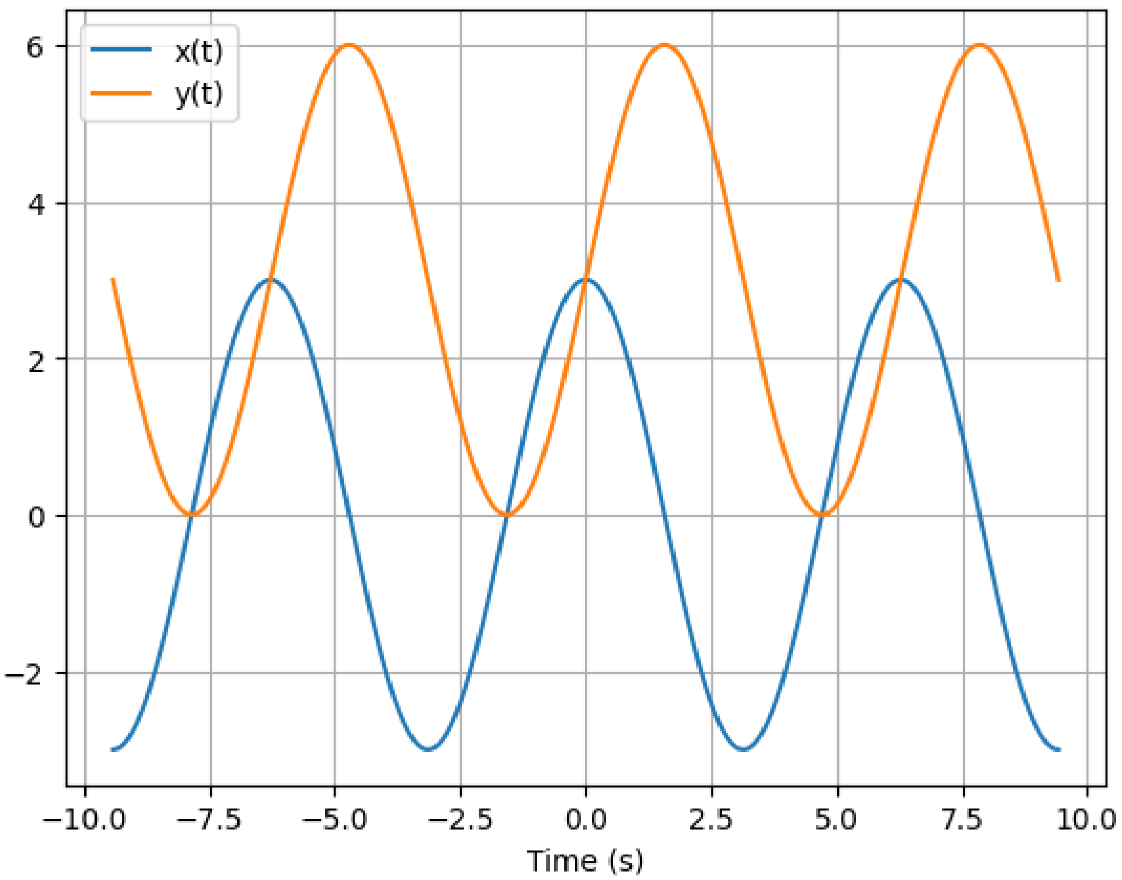
Tema 2: Operación de la variable dependiente/independiente

(1) Graficar la señal original y la resultante de la operación descrita en el ejercicio. Si la señal es compleja, graficar parte real y parte imaginaria.

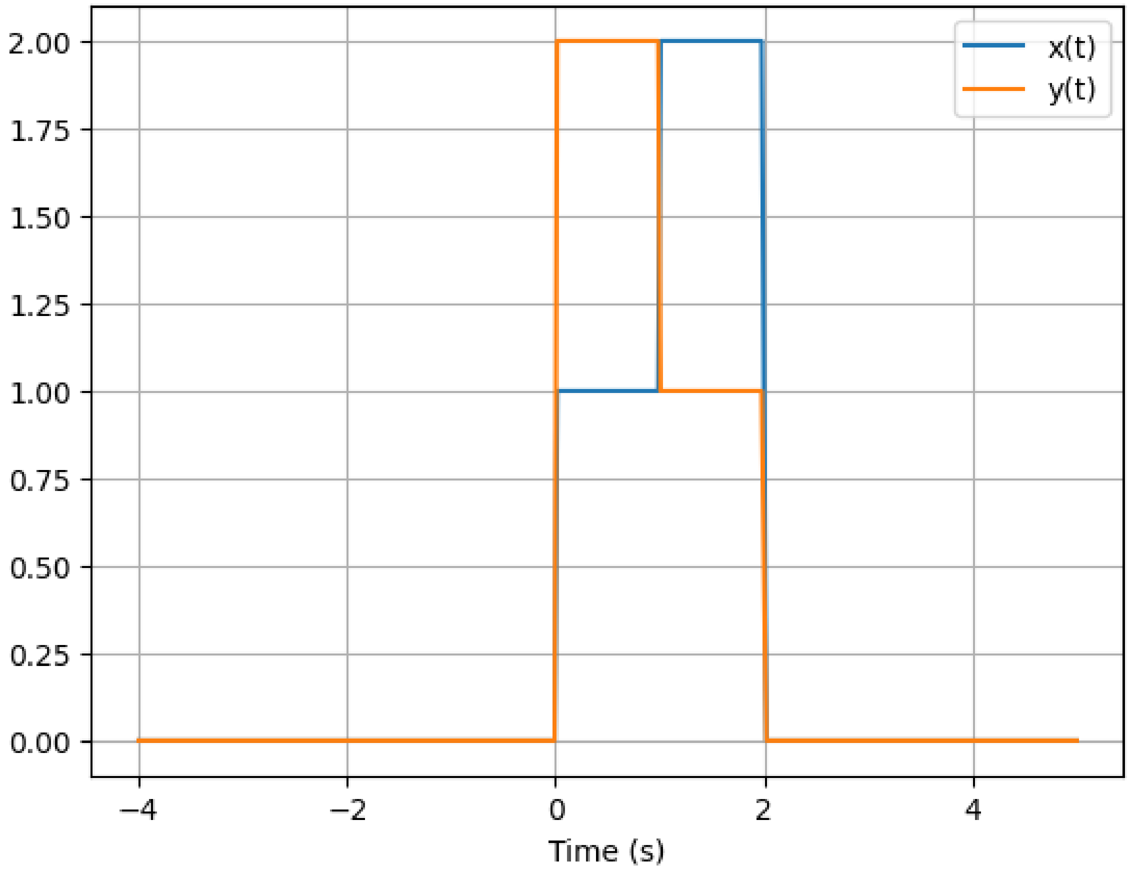
1. $y(t) = 3 \cdot x(t)\ con\ x(t) = 2sen(t)$



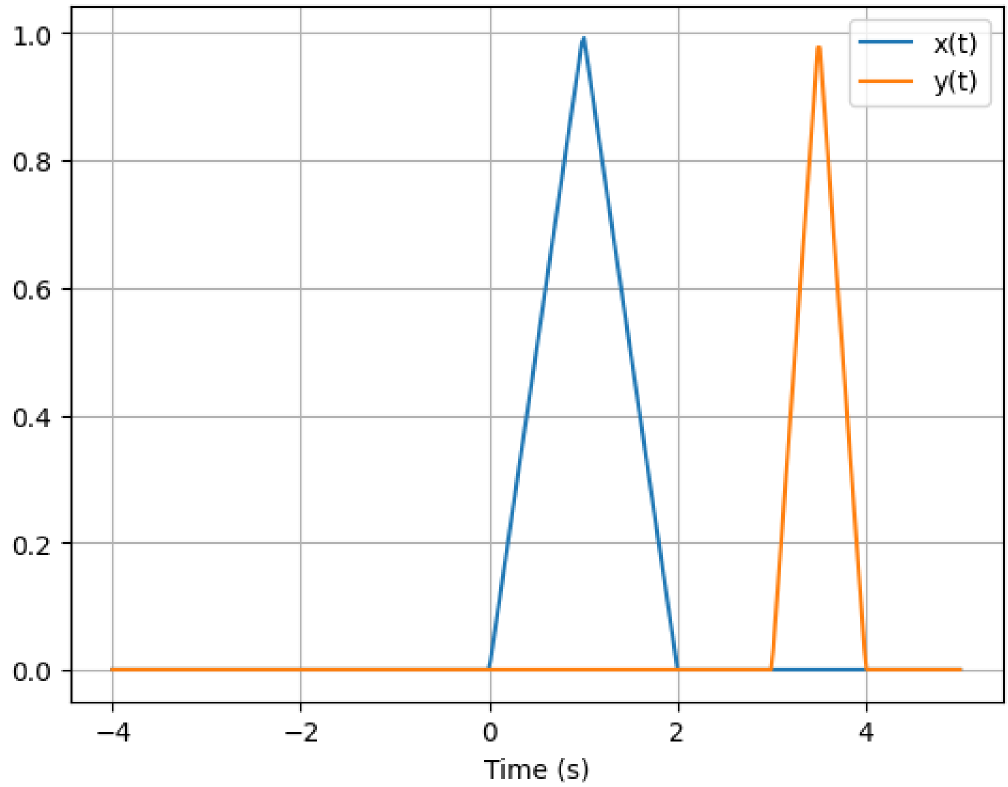
2. $y(t) = x(t - \frac{\pi}{2}) + 3\ con\ x(t) = 3cos(t)$



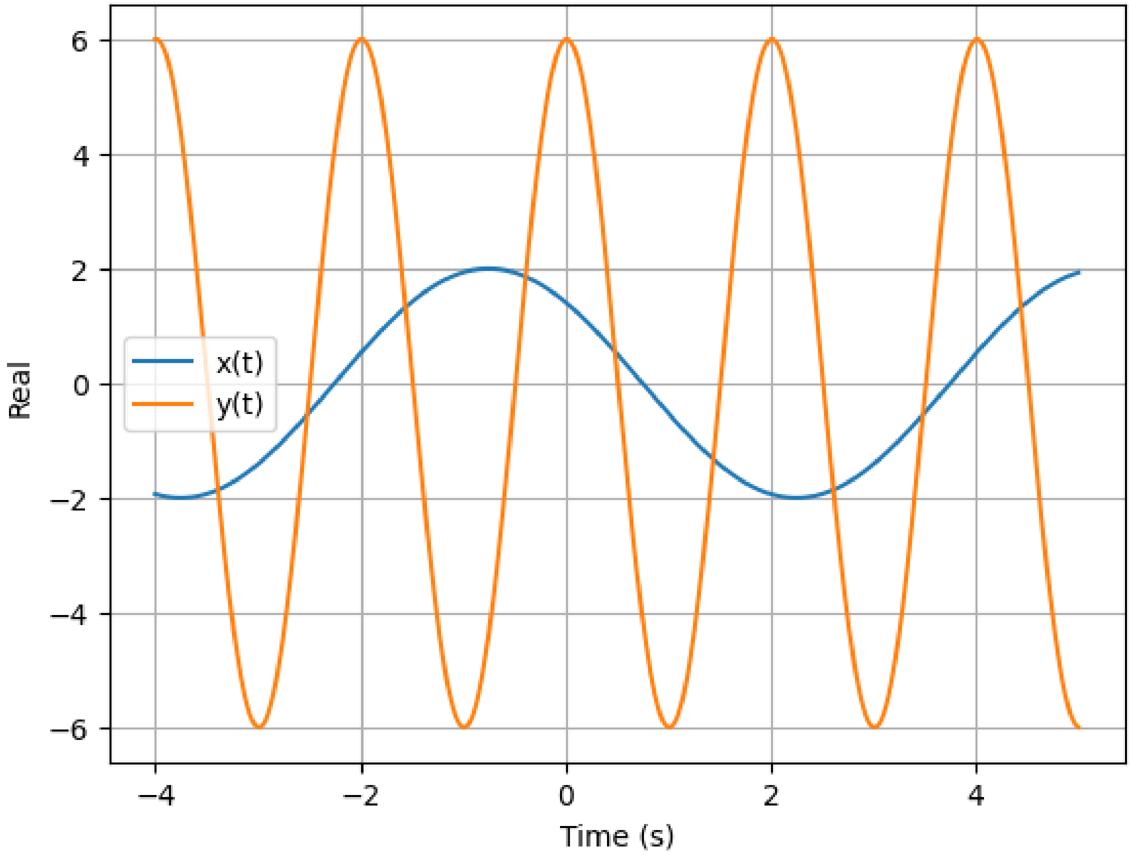
3. $y(t) = x(2-t)$ *con* $x(t) = \mu(t) + \mu(t-1) - 2\mu(t-2)$

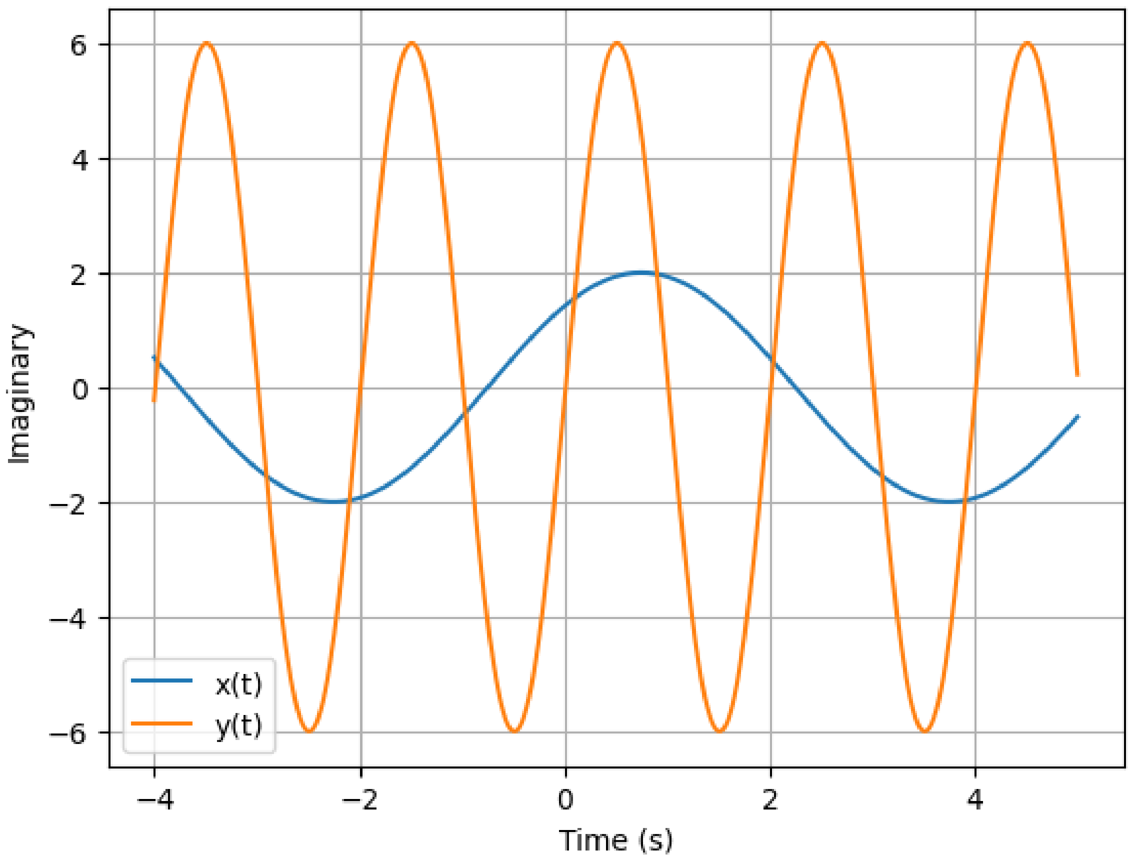


4. $y(t) = x(-2t+3)$ *con* $x(t) = t \cdot [\mu(t) - \mu(t-1)] + (-t+2) \cdot [\mu(t-1) - \mu(t-2)]$



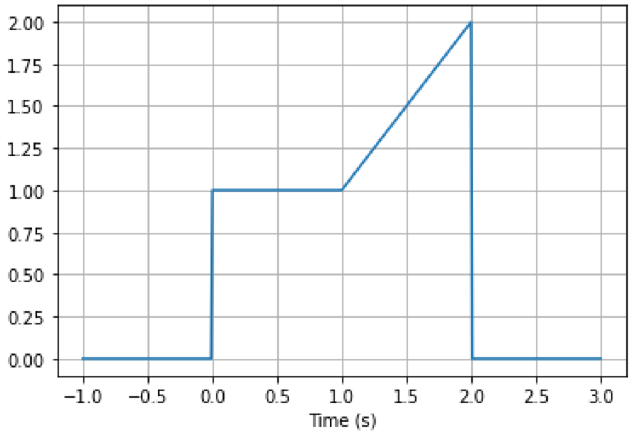
5. $y(t) = 3\ x(3t - \frac{pi}{4})\ con\ x(t) = 2e^{j\left(\frac{\pi}{3}t + \frac{\pi}{4}\right)}$



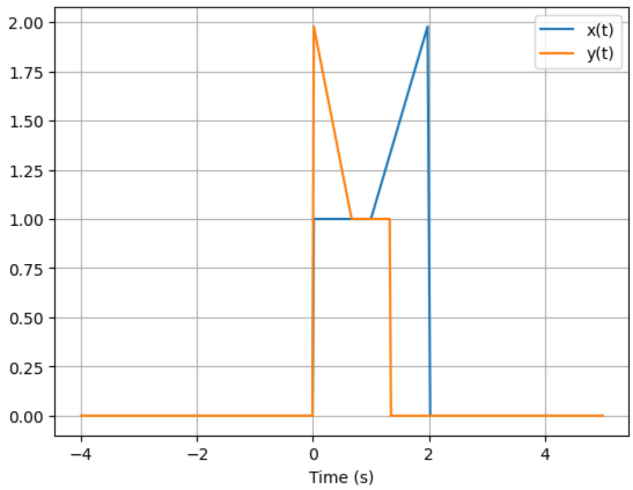


(2) Describir la señal de la imagen a través de su expresión matemática. Aplicar la operación correspondiente y graficar

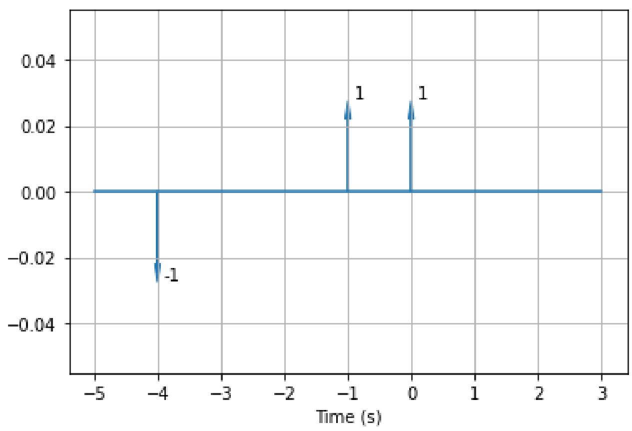
1. $y(t) = x(-\frac{3}{2}t + 2)$



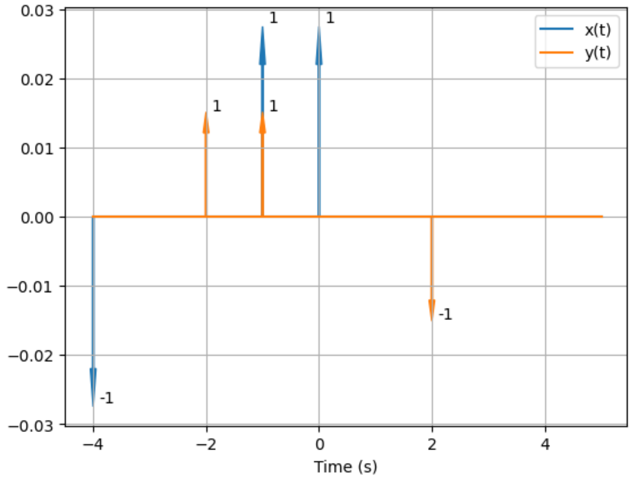
$x(t) = u(t) - u(t-1) + t * (u(t-1) - u(t-2))$



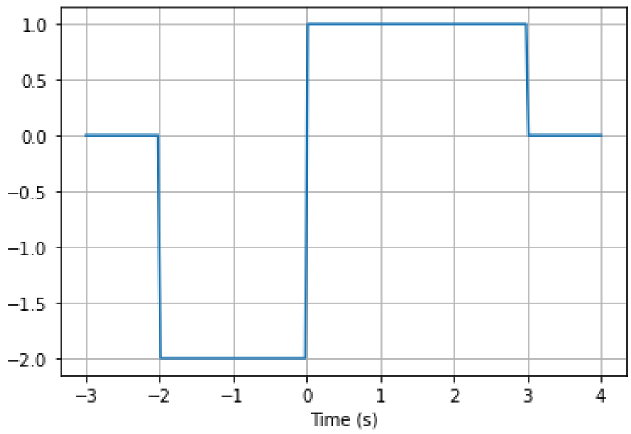
2. $y(t) = x(-t - 2)$



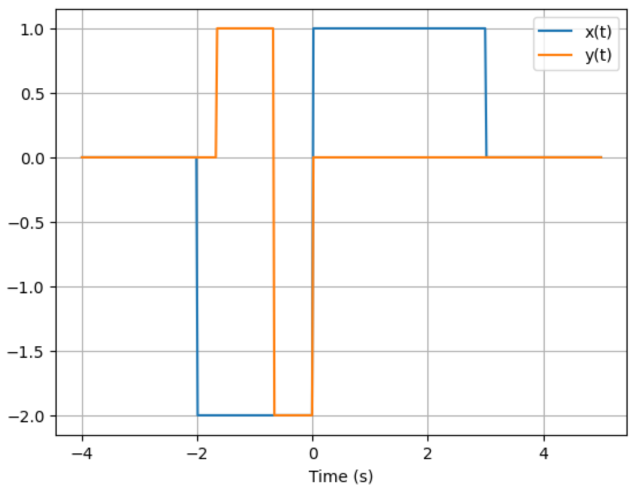
$x(t) = -\delta(t+4) + \delta(t+1) + \delta(t)$



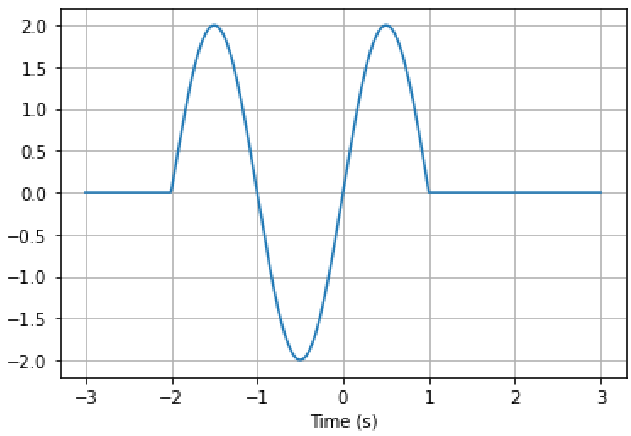
3. $y(t) = x(-2 + 3t)$



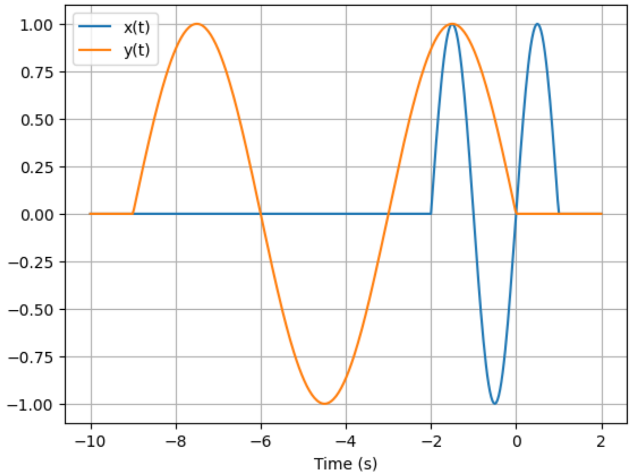
$$x(t) = -2u(t + 2) + 3u(t) - u(t - 3)$$



4. $y(t) = x(1 + \frac{1}{3}t)$



$x(t) = sen(\pi t) \cdot (u(t+2) - u(t-1))$



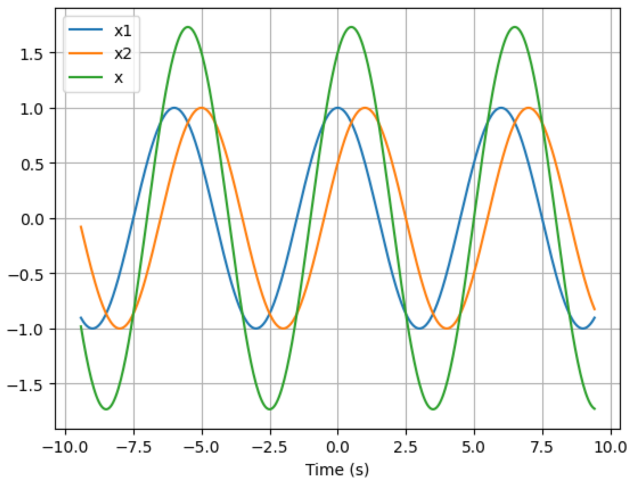
Tema 3: Periodicidad. Energía y Potencia de una señal

(1) Aplicar las operaciones correspondientes descritas en el ejercicio. Determinar si la señal es periódica o no y si lo es, expresar su período fundamental. Graficar las señales y sus resultados.

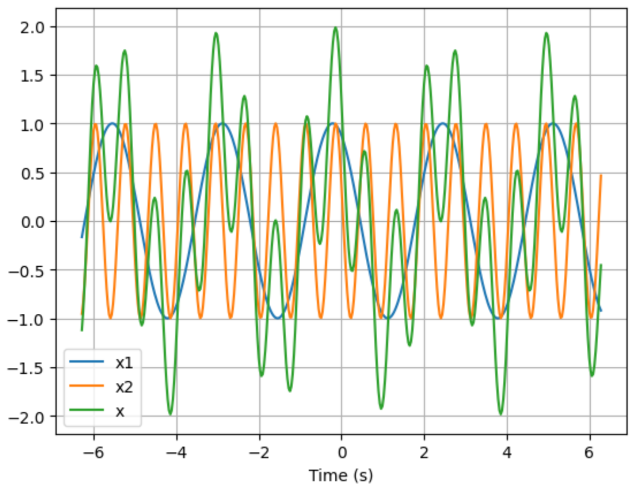
1. Calcular el período de $x(t) = cos\,(\omega_x t + \phi_x) + cos\,(\omega_y t + \phi_y)$ siendo:

ω_x	ϕ_x	ω_y	ϕ_y
$\pi/3$	0	$\pi/3$	$-\pi/3$
$3\pi/4$	$1/2$	$11\ \pi/4$	$3\pi/8$
$3\pi/2$	$1/5$	$5\ \pi/4$	$3\pi/4$
$3/4$	$1/2$	$3/4$	1
$3\pi/4$	$\pi/2$	$2/3$	1

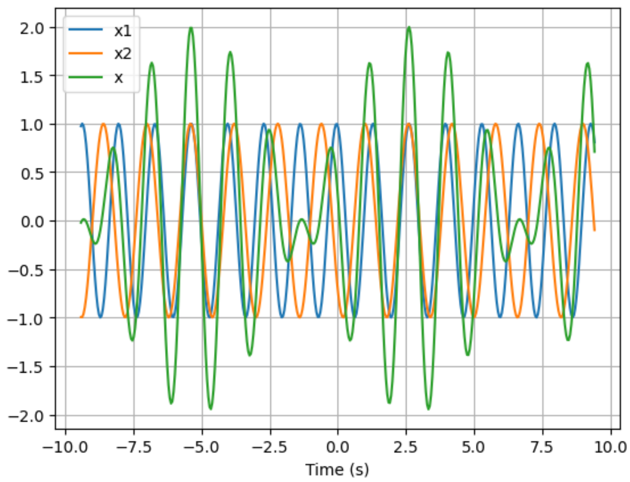
- $x(t) = cos(\frac{\pi}{3}t) + cos(\frac{\pi}{3}t - \pi/3)$, $T_1 = 6$, $T_2 = 6$, $T_0 = 6$



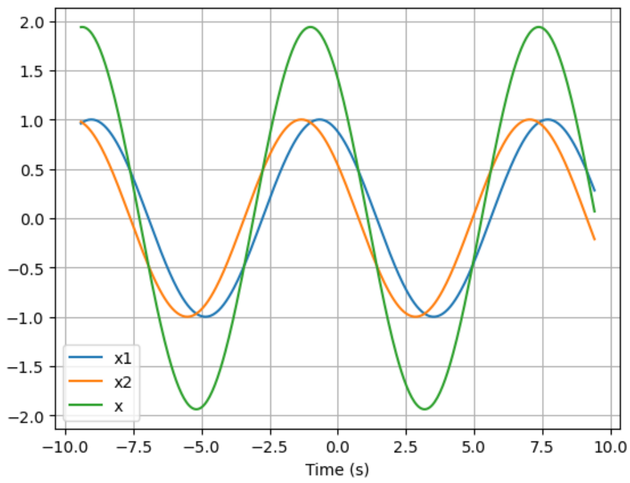
- $x(t) = cos(\frac{3\pi}{4}t + \frac{1}{2}) + cos(\frac{11\pi}{4}t + \frac{3\pi}{8})$, $T_1 = 8/3$, $T_2 = 8/11$, $T_0 = 8$



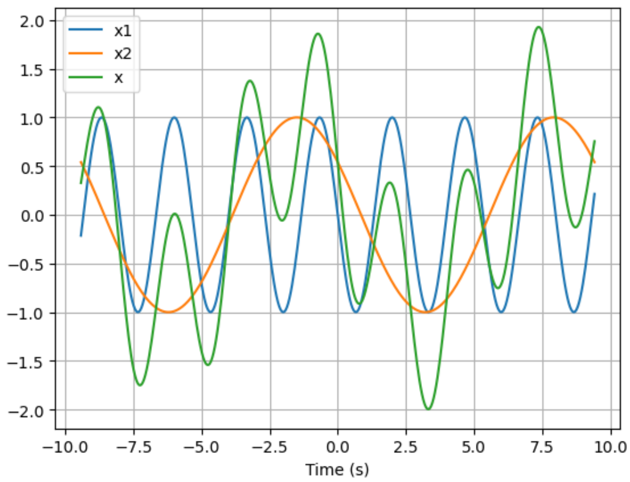
- $x(t) = cos(\frac{3\pi}{2}t + \frac{1}{5}) + cos(\frac{5\pi}{4}t + \frac{3\pi}{4})$, $T_1 = 4/3$, $T_2 = 8/5$, $T_0 = 8$



- $x(t) = cos(\frac{3}{4}t + \frac{1}{2}) + cos(\frac{3}{4}t + 1)$, $T_1 = 8/3$, $T_2 = 8/3$, $T_0 = 8/3$



- $x(t) = cos(\frac{3\pi}{4}t + \frac{\pi}{2}) + cos(\frac{2}{3}t + 1$, $T_1 = 8/3$, $T_2 = 3\pi$, NO ES PERIODICA

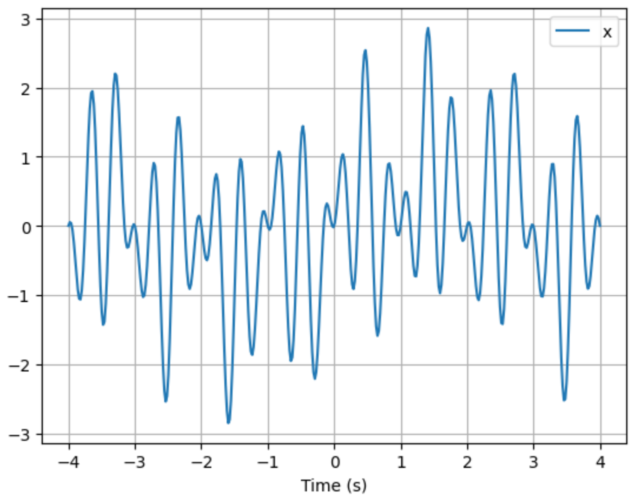


2. Siendo:

$$x(t) = cos(2\pi t/3) + 2sen(16\pi t/3) \qquad y(t) = sen(\pi t)$$

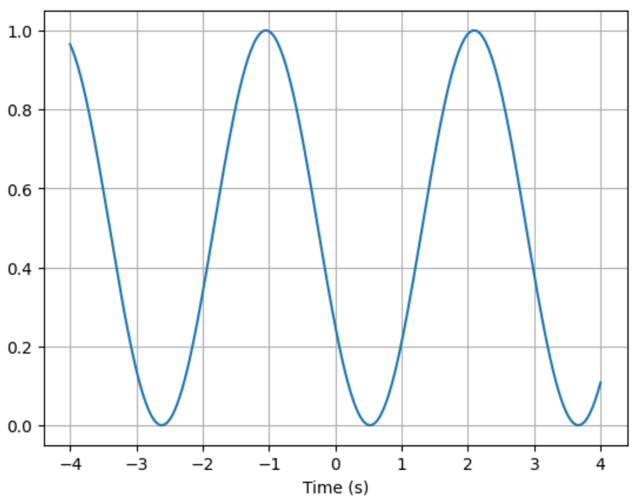
Determinar si $z(t) = x(t){\cdot}y(t)$ es periódica y si así es, cuál es su período fundamental. Identidades necesarias: $2sen(\alpha)cos(\beta) = sen(\alpha - \beta) + sen(\alpha + \beta)$ y $2sen(\alpha)sen(\beta) = cos(\alpha - \beta) - cos(\alpha + \beta)$

$z(t) = \frac{1}{2}sen(\frac{\pi}{3}t) + sen(\frac{5\pi}{3}t) + cos(\frac{13\pi}{3}t) - cos(\frac{19\pi}{3}t)$, $T_0 = 6$



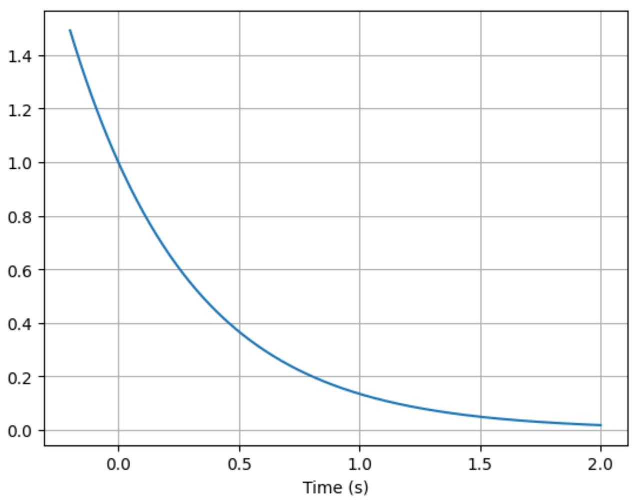
3. Determinar si $x(t) = (sen(t - \pi/6))^2$ es periódica y su período fundamental. Identidad necesaria: $2sen^2(\alpha) = 1 - cos(2\alpha)$

$x(t) = \frac{1}{2}\left[1 - cos(2t - \frac{2}{6}\pi)\right]$, $T_0 = \pi$

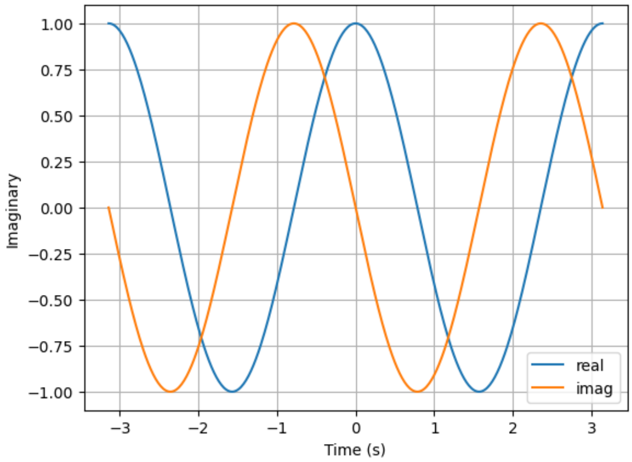


4. Determinar el período de las siguientes señales compuestas por exponenciales:

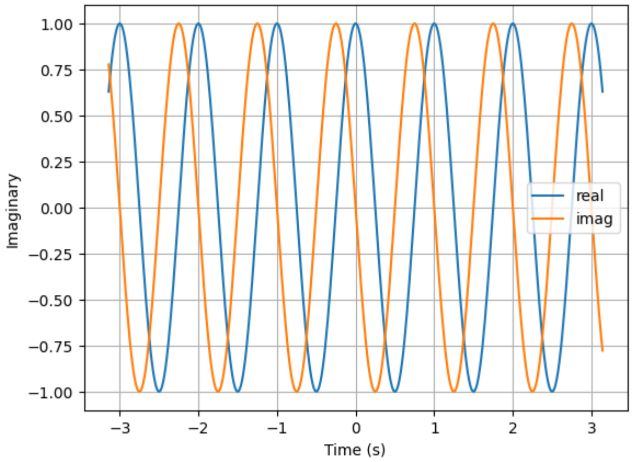
- $x(t) = e^{-2t}$, No es periódica



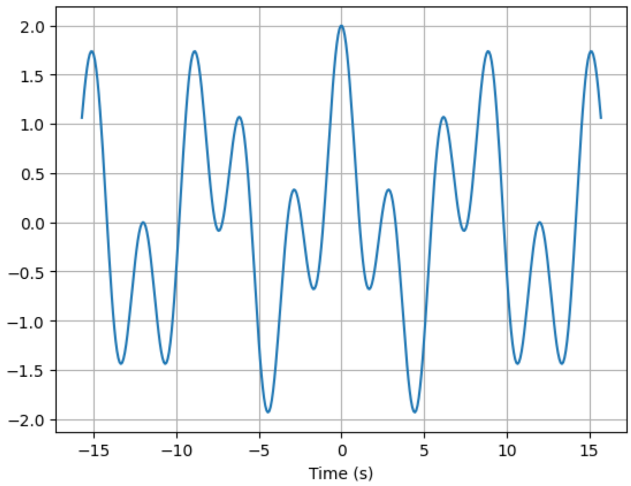
- $x(t) = e^{-2jt}$, $T_0 = \pi$



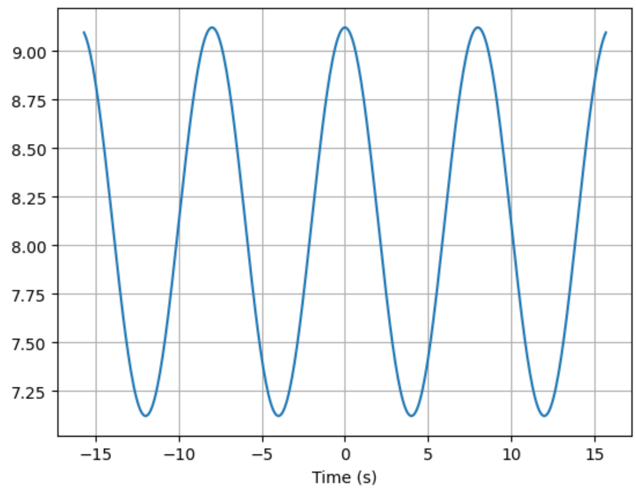
- $x(t) = e^{j2\pi t}$, $T_0 = 1$



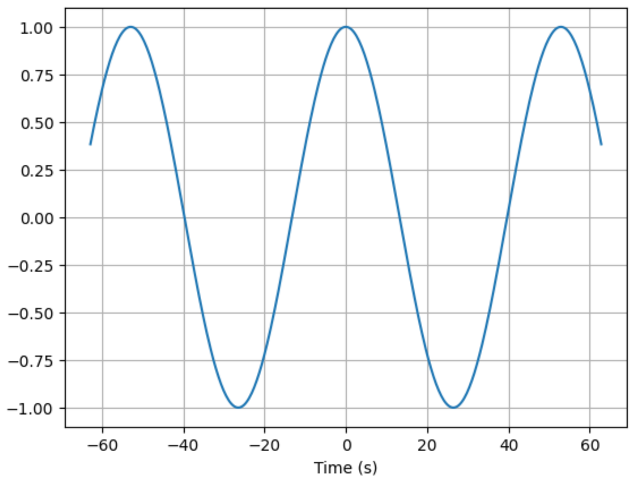
- $x(t) = e^{-j\pi t/4} + e^{j2\pi t/3}$, $T_0 = 24$



- $x(t) = e^{-j\pi t/4} + e^{2\pi/3}$, $T_0 = 8$



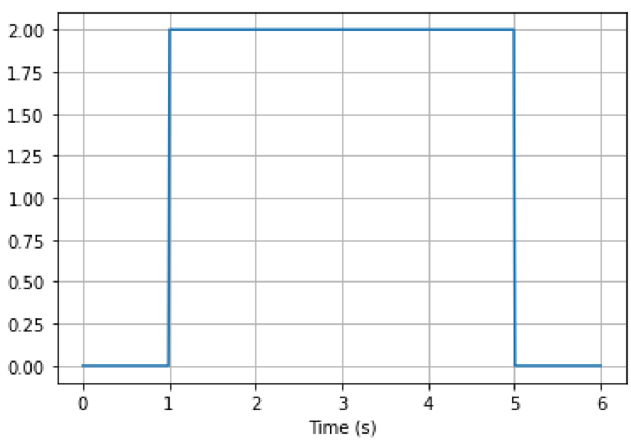
- $x(t) = e^{-j\pi t/4} \cdot e^{j2t/3}$, $T_0 = 52.92$



- $x(t) = e^{-j\pi t/4} + e^{j2t/3}$, No es periódica

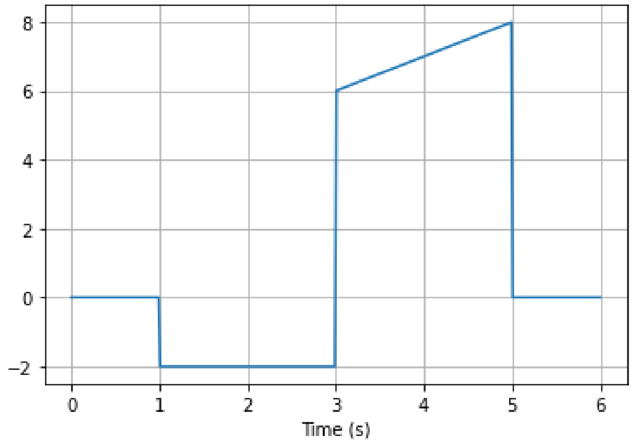
(2) Aplicar las operaciones correspondientes descritas en el ejercicio. Calcular la energía y potencia de la señal en un período (si es periódica) y en el infinito. Determinar si es una señal de energía.

1.



$E_\infty = 16$
$P_{1;5} = 4$
$P_\infty = 0$
No es periódica. Es una señal de energía.

2.



$E_\infty = 64.67$
$P_{1;5} = 16.17$
$P_\infty = 0$
No es periódica. Es una señal de energía.

3. $x(t) = e^{(-2+j2\pi)t}$
$E_\infty = \infty$
$P_\infty = \infty$
No es periódica, tampoco es una señal de energía.

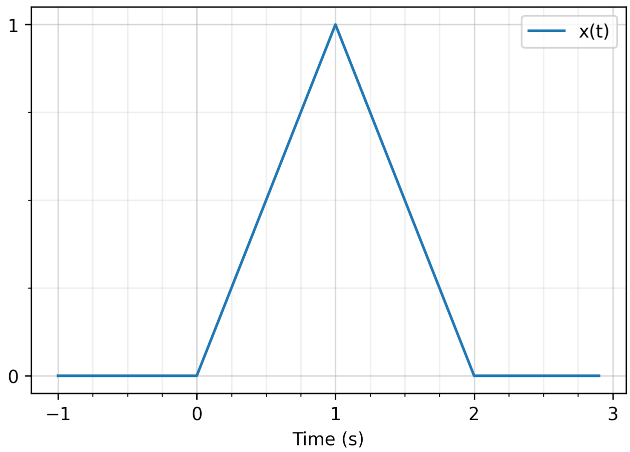
4. $x(t) = e^{-2t}$
$E_\infty = \infty$
$P_\infty = \infty$
No es periódica, tampoco es una señal de energía.

5. $x(t) = e^{-2t} \cdot \mu(t)$
$E_\infty = \frac{1}{4}$
$P_\infty = 0$
No es periódica, es una señal de energía.

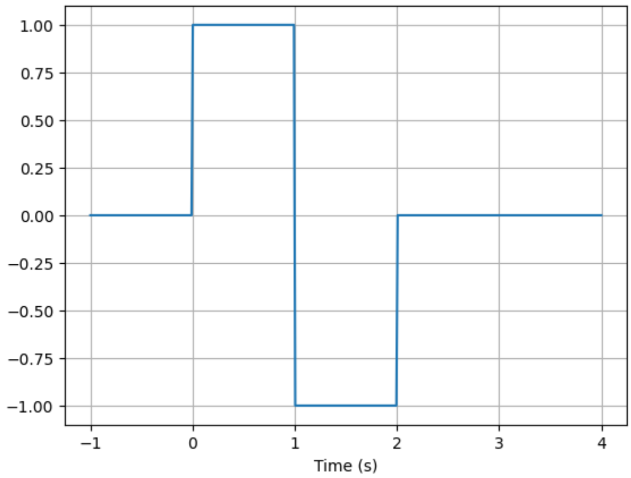
Tema 4: Derivadas e Integrales de una señal

(1) Calcular la derivada de las siguientes señales de manera analítica y gráfica.

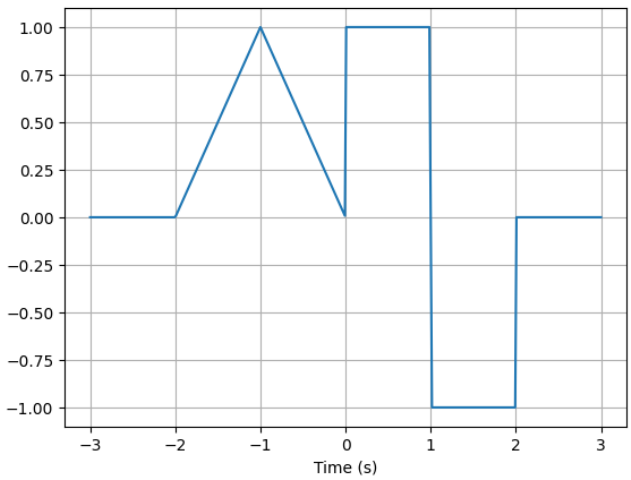
1.



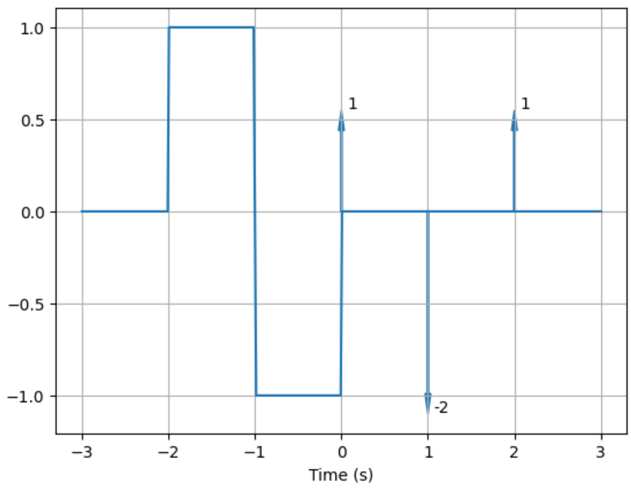
$x(t) = t \cdot [u(t) - u(t-1)] + (-t+2) \cdot [u(t-1) - u(t-2)]$
$\frac{\partial}{\partial t} x(t) = u(t) - 2u(t-1) + u(t-2)$



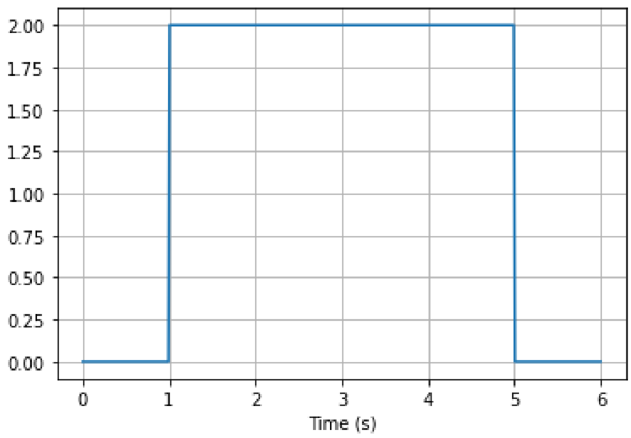
2.



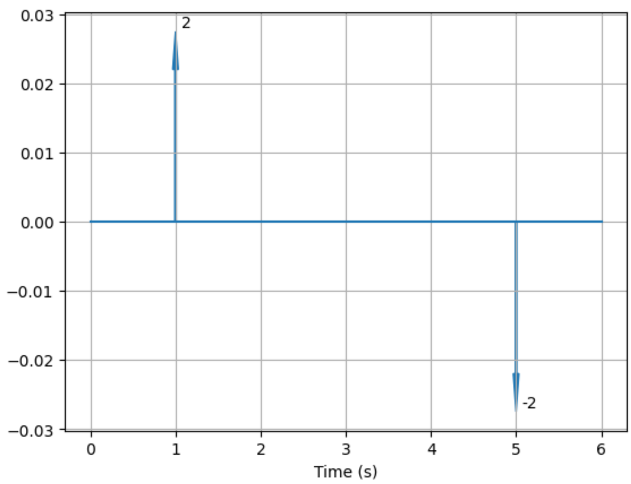
$x(t) = (t+2) \cdot (u(t+2) - u(t+1)) + (-t) \cdot (u(t+1) - u(t)) + u(t) - 2u(t-1) + u(t-2)$
$\frac{\partial}{\partial t} x(t) = u(t+2) - 2u(t+1) + u(t) + \delta(t) - 2\delta(t-1) + \delta(t-2)$



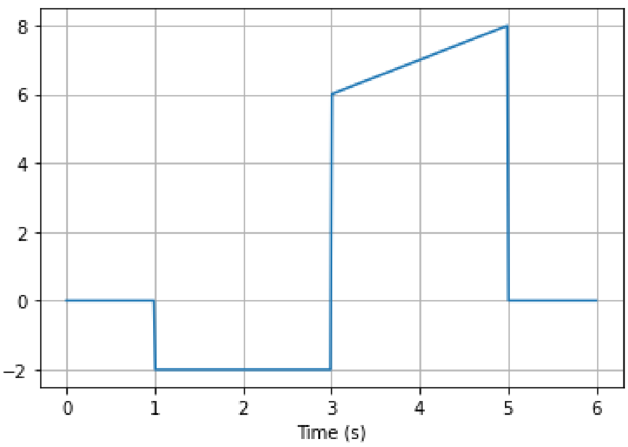
3.



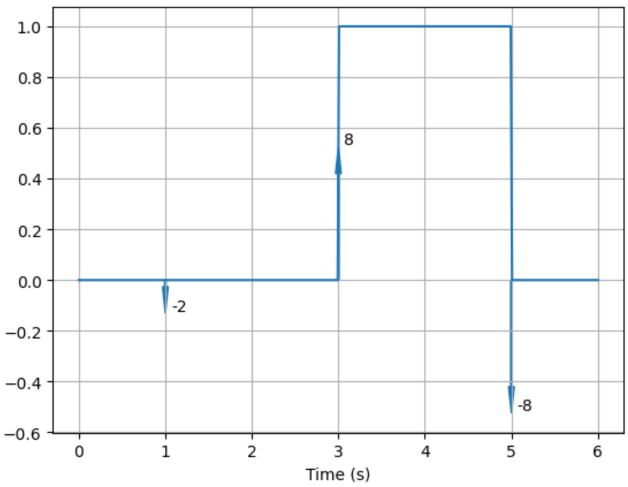
$$x(t) = 2 * u(t-1) - 2 * u(t-5)$$
$$\frac{\partial}{\partial t} x(t) = 2 * \delta(t-1) - 2 * \delta(t-5)$$



4.

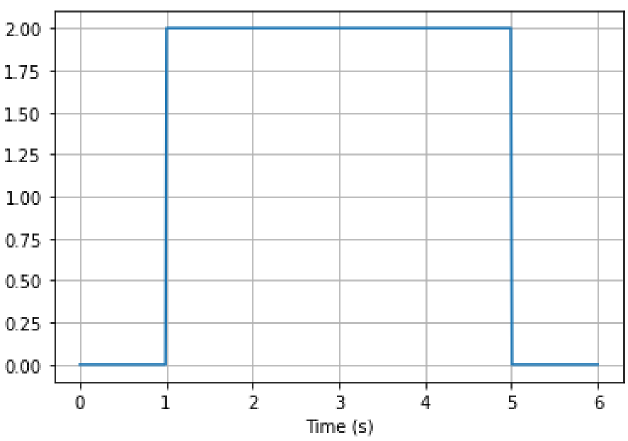


$x(t) = -2(u(t-1) - u(t-3)) + (t+3)(u(t-3) - u(t-5))$
$\frac{\partial}{\partial t} x(t) = -2\delta(t-1) + +8\delta(t-3) + u(t-3) - u(t-5) - 8\delta(t-5)$

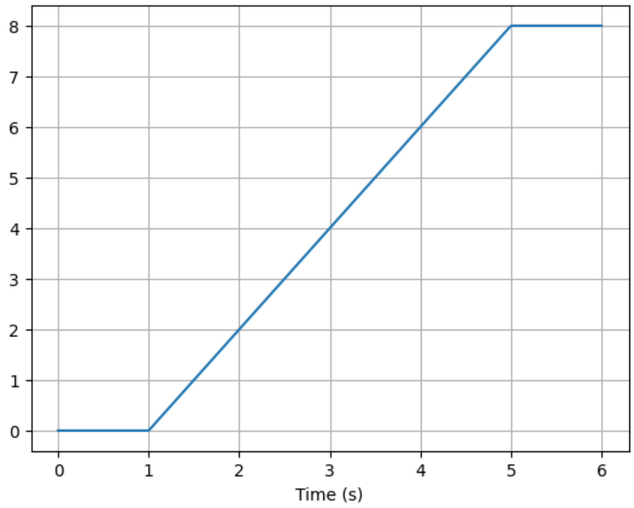


(2) Calcular la integral de las siguientes funciones

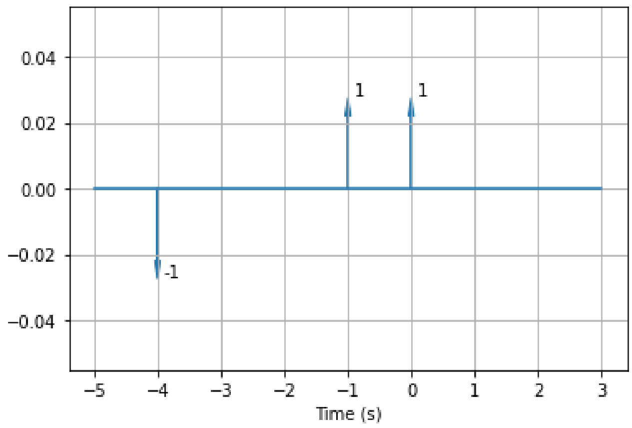
1.



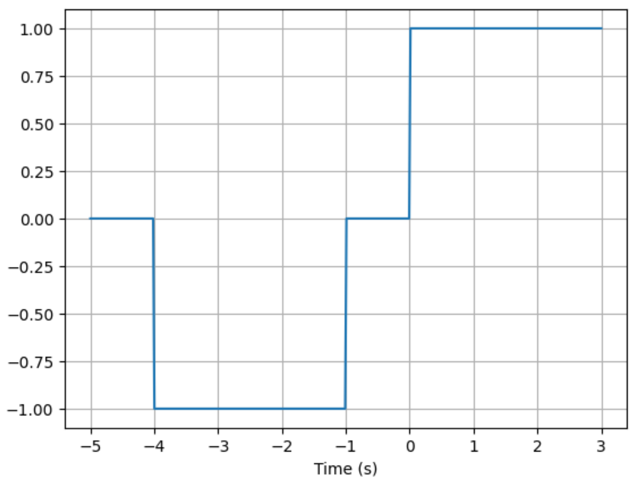
$x(t) = 2(u(t-1) - u(t-5))$
$\int x(t)dt = (2t-2)u(t-1) - (2t-10)u(t-5)$



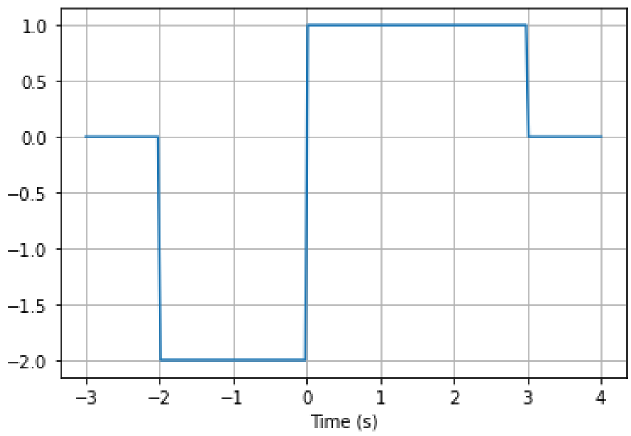
2.



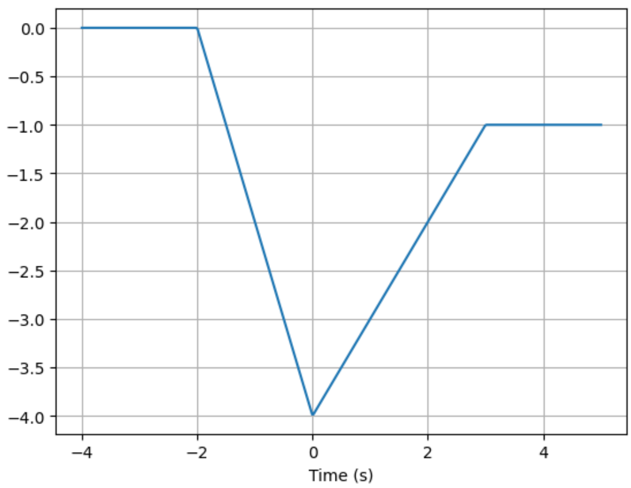
$$x(t) = -\delta(t+4) + \delta(t+1) + \delta(t)$$
$$\int x(t)dt = -u(t+4) + u(t+1) + u(t)$$



3.



$x(t) = -2(u(t+2) - u(t)) + u(t) - u(t-3)$
$\int x(t)dt = (-2t-4)(u(t+2) - u(t)) + (-4+t) * (u(t) - u(t-3)) - u(t-3)$



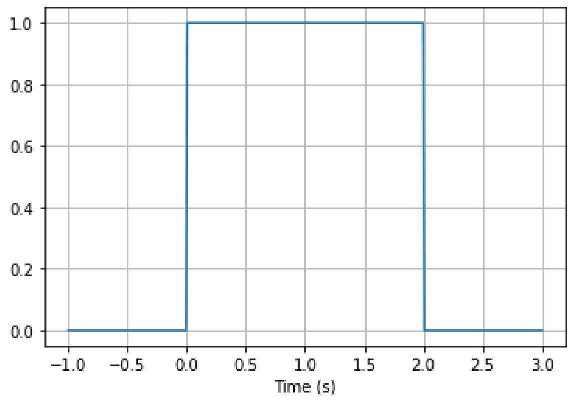
Tema 5: Sistemas

(1) Indicar si los sistemas expresados cumplen o no las propiedades de la tabla

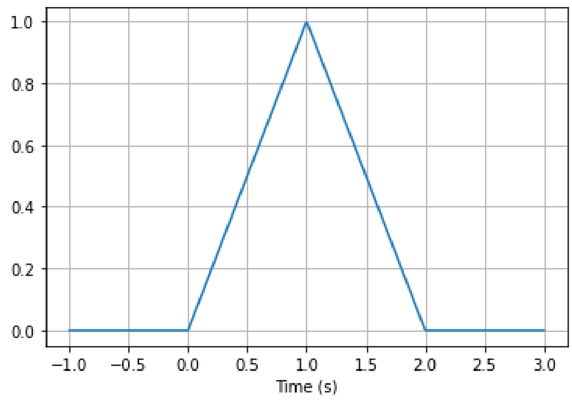
y(t)	**Memoria**	**Lineal**	**Invariante en el tiempo**	**Causal**	**Invertible**	**Estable**
$(2 + sen(t))x(t)$	No	Si	No	Si	No	Si
$x(2t)$	Si	Si	No	No	Si	Si
$\int_{-\infty}^{t} x(\tau)d\tau$	Si	Si	Si	Si	Si	No
$\frac{\partial x(t)}{\partial t}$	Si	Si	Depende	Si	No	No
$e^{x(t)}$	No	No	Si	Si	Si	Si
$x(t) \cdot x(t-1)$	Si	No	Si	Si	No	Si
$sen(6t) \cdot x(t)$	No	Si	No	Si	No	Si
$x(t-1) - 2x(t-4)$	Si	Si	Si	Si	Si	Si

(2) Sea un sistema SLIT cuya respuesta a la señal $x_1(t)$ de la figura es la señal $y_1(t)$, determinar la respuestas de las entradas $x_2(t)$ y $x_3(t)$

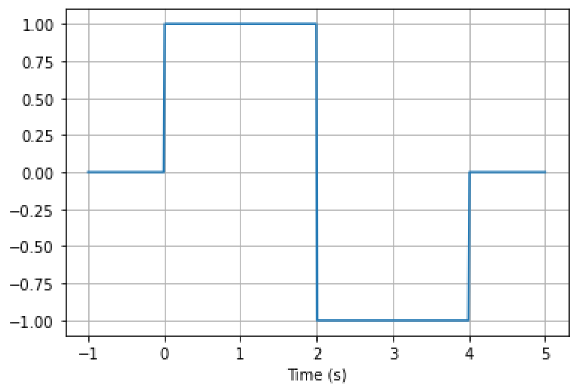
1.



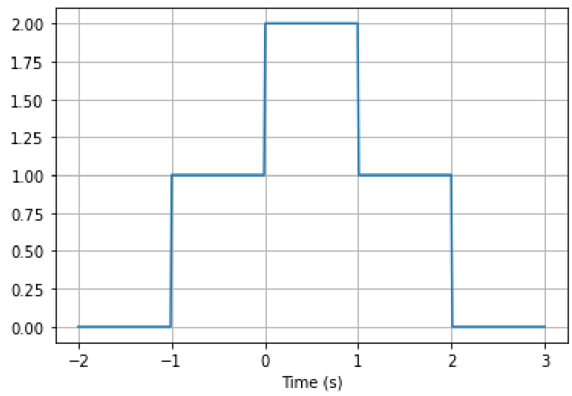
(a) x1(t)



(b) y1(t)

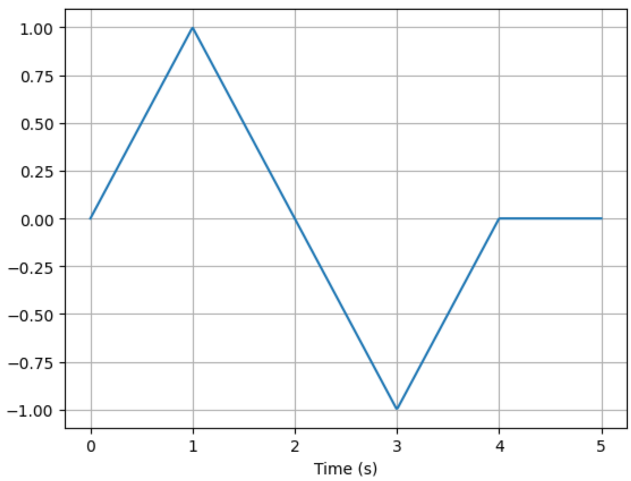


(a) x2(t)

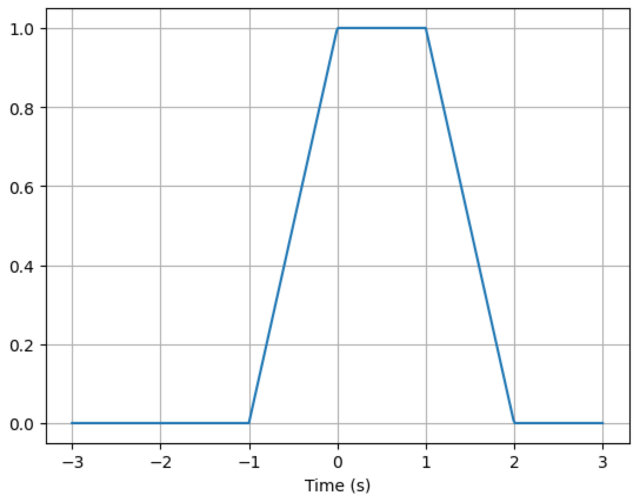


(b) x3(t)

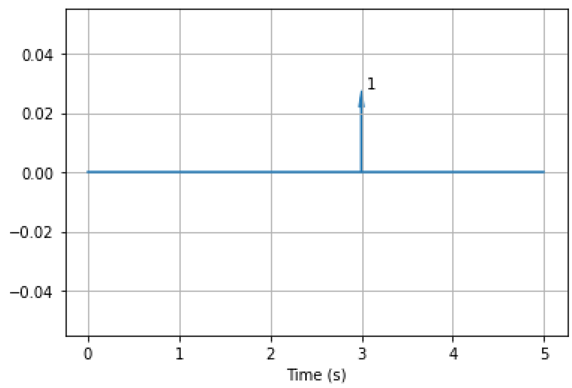
$$y_2 = t(u(t)-u(t-1))+(-t+2)(u(t-1)-u(t-2))+(-t+2)(u(t-2)-u(t-3))+(t-4)(u(t-3)-u(t-4))$$



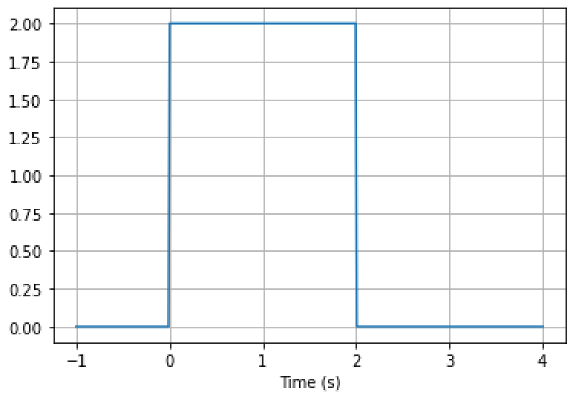
$$y_3 = (t+1)(u(t+1)-u(t))+(u(t)-u(t-1))+(-t+2)(u(t-1)-u(t-2))$$



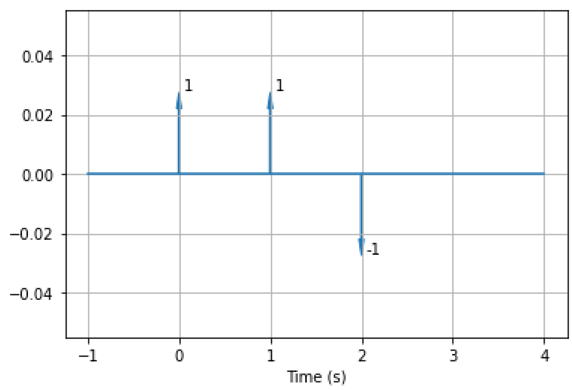
2.



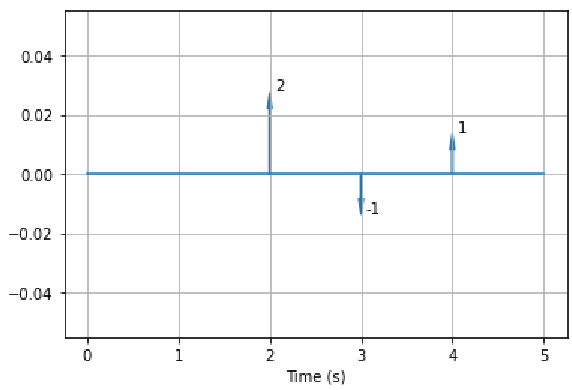
(a) x1(t)



(b) y1(t)

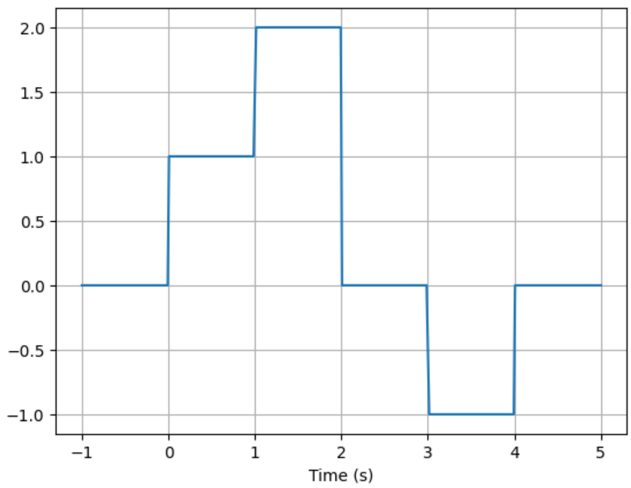


(a) x2(t)



(b) x3(t)

$$y_2(t) = u(t) + u(t-1) - u(t-2) - u(t-3) - u(t-2) + u(t-4)$$



$y_3(t) = 2u(t-2) - u(t-3) - 2u(t-4) + u(t-4) + u(t-5) - u(t-6)$